Hesai AT512 for Emergency Brake Assist

Lidar performance – statistical modelling (Exercise 3)

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Task and requirements

Question: does a high-end automotive lidar on a highway car fulfil the EBA requirements?

- detection range **250 m**
- at least **5 returns per object**

Our criterion: $P(\text{at least 5 returns}) \geq 99\,\%$ in a single frame, at 250 m, for dark (10 %) objects.

Objects (front view, W × H)

pedestrian	0.5×1.8 m
motorcycle	0.8×1.4 m
car	1.8×1.5 m
truck	2.5×3.5 m
tyre / lost cargo	0.5×0.5 m

The lidar: Hesai AT512

range	300 m @ 10 % refl. (400 m max)
resolution	$0.05^\circ \times 0.05^\circ$
field of view	$120^\circ \times 25.6^\circ$
points	$12.3\,\mathrm{M/s} = 2400 \times 512$ @ 10 Hz

Why this one

- automotive-grade, long-range series product
- very fine angular grid → many beams on small, far objects
- 120° covers adjacent lanes

Assumption: the 300 m spec means 90 % detection probability (not stated by Hesai).

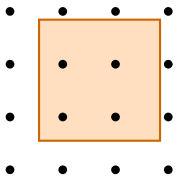
Model, step 1: how many beams hit the object?

Beam spacing at distance R: $s = R\,\Delta\theta$
(250 m $\rightarrow$ 0.22 m)

$$n = \frac{W}{R\,\Delta\theta_h} \cdot \frac{H}{R\,\Delta\theta_v} \propto \frac{1}{R^2}$$

The grid has a random offset: a side of $L = 2.29$ spacings gets 2 beams (71 %) or 3 beams (29 %).

Car @ 250 m: $8.3 \times 6.9 \approx 57$ beams
Tyre @ 250 m: $2.3 \times 2.3 \rightarrow 4$, 6 or 9 beams



0.5 m tyre, beams every 0.22 m
this offset: $2 \times 2 = 4$ hits

Model, steps 2–3: returns are a binomial random variable

Per-beam detection probability – echo power $\propto \rho/R^2$:

$$p(R,\rho) = 1 - \exp\Big(-C\,\frac{\rho}{R^2}\Big), \qquad C \text{ from the datasheet: } p(300\,\text{m}, 10\,\%) = 0.9$$

$\Rightarrow\ p(250\,\text{m}, 10\,\%) = 0.964$

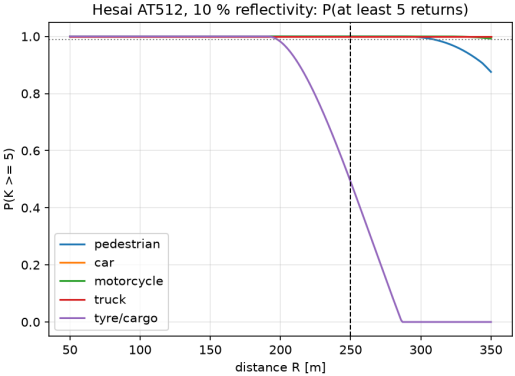
Number of returns: n independent beams, each returns with p

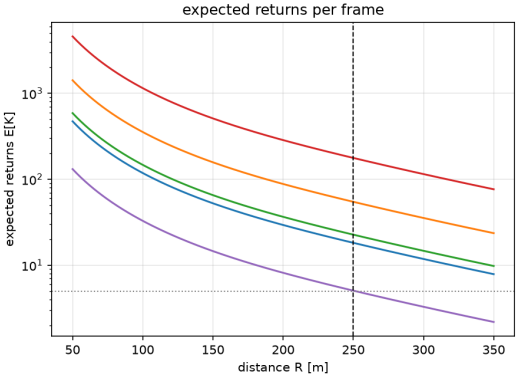
$$K \sim \text{Binomial}(n,p), \qquad P(K \geq 5) = 1 - \sum_{k=0}^{4} \binom{n}{k} p^k (1-p)^{n-k}$$

averaged over the possible beam counts n (random grid offset).

Verdict: largest R with $P(K \geq 5) \geq 99\,\%$ – is it $\geq 250\,\text{m}$?

Results (10 % reflectivity)





Results

object (10 %)	$E[K]$ @ 250 m	$R_{99\%}$	
pedestrian	18	308 m	pass
motorcycle	23	353 m	pass
car	55	$\geq$ 400 m	pass
truck	177	$\geq$ 400 m	pass
tyre / cargo	5.1	197 m	fail

Tyre @ 250 m: $P(K \geq 5) = 0.49$

- half the time only $2 \times 2 = 4$ beams hit
- same result at 50 % reflectivity

$\Rightarrow$ limited by **resolution**, not by sensitivity

Limitations and conclusion

Assumptions

- flat, frontal targets (no windows, no curvature)
- independent beams – optimistic in rain or with a dirty sensor
- 90 % behind the 300 m spec is assumed
- single frame; 34 frames remain before braking at 130 km/h (stopping distance $\approx$ 127 m)

Conclusion

- **AT512 fulfils EBA** for all vehicles and pedestrians at 250 m, even at 10 % reflectivity
- small debris ($\sim$0.5 m) is reliable only up to $\approx$ 200 m – a resolution limit